Ex.No: 03 IMPLEMENTATION OF LOG MONITORING FOR FORENSIC READINESS

Date:

AIM:

To implement log monitoring for forensic readiness.

SOFTWARE/EQUIPMENTS REQUIRED:

- Windows 10/11
- Windows Powershell

THEORY:

Log monitoring is an important component of digital forensic readiness because system logs contain records of activities performed on a computer. These records may include login attempts, authentication failures, system activities, timestamps, and other security-related events. Windows maintains security-related events in the Windows Security Event Log. By continuously monitoring these events, suspicious activities can be identified and recorded for further investigation.

TOOL USED:

PowerShell is used to access and monitor the Windows Security log. Event ID 4625 is used to identify failed login attempts. Multiple failed login attempts are treated as suspicious activity for the purpose of this experiment. The identified events are then exported to a CSV file so that they can be preserved and analyzed as potential forensic evidence.

ALGORITHM:

1. Start the program.
2. Open Windows PowerShell with administrator privileges.
3. Access the Windows Security Event Log.
4. Generate a few failed login attempts on the local computer.
5. Search the Security log for Event ID 4625.
6. Retrieve and count the detected failed login events.
7. If the count is three or more, display a suspicious activity alert.
8. Create a folder to preserve forensic evidence.
9. Export the detected security events to a CSV file.
10. Verify the exported file and examine its contents.
11. Preserve the evidence.
12. End

PROCEDURE:

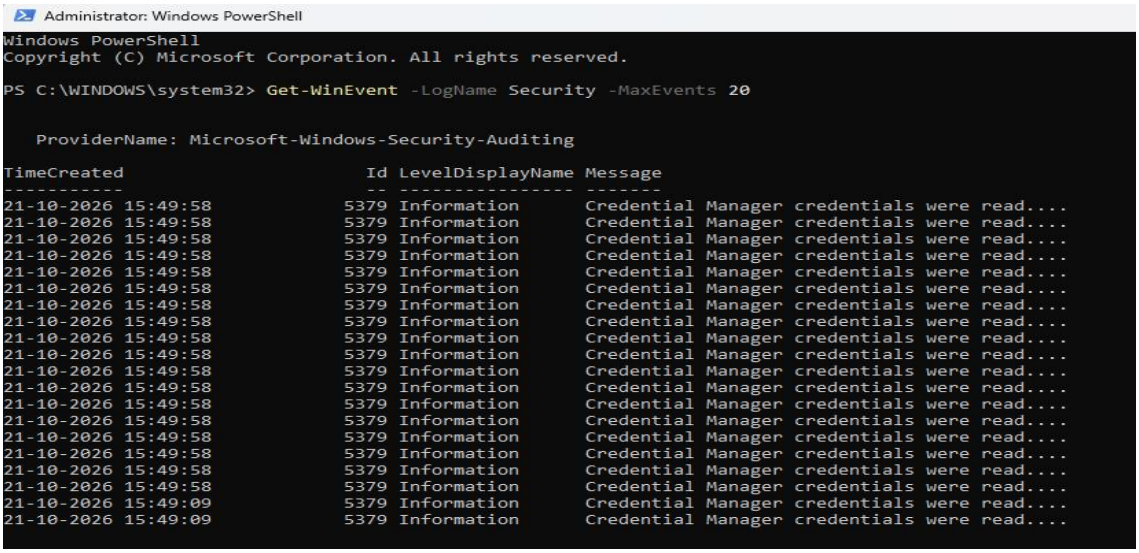
Step 1: Open PowerShell as Administrator

1. Click the Windows Start button.
2. Search for `PowerShell`.
3. Right-click Windows PowerShell.
4. Select Run as administrator.

Step 2: Check the Security Log

1. Copy and run this command:

Get-WinEvent -LogName Security -MaxEvents 20



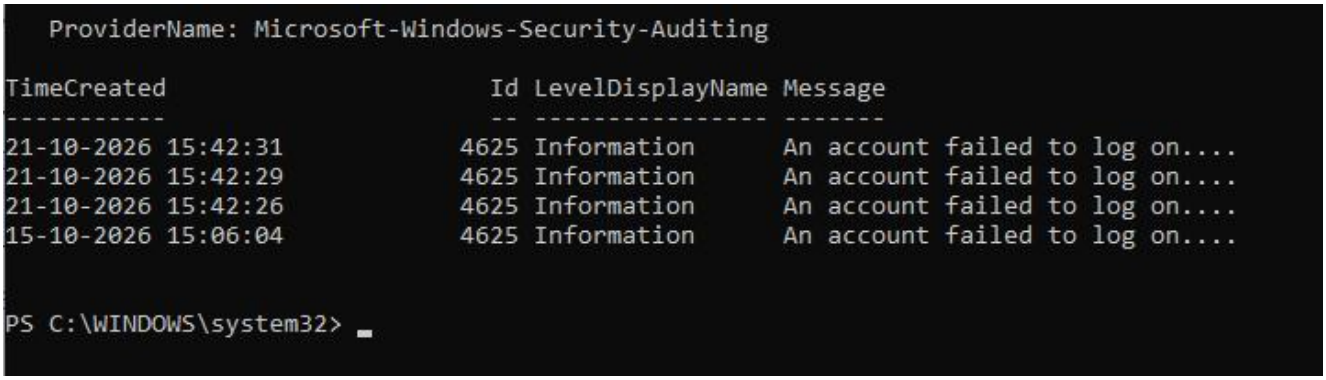
Step 3: Generate failed login attempts

1. Press Windows + L to lock the computer.
2. Enter an incorrect password a few times.
3. Enter the correct password to unlock the computer.

Step 4: Find failed login events

Return to PowerShell and run:

```
Get-WinEvent -FilterHashtable @{
    LogName = 'Security'
    Id = 4625
} -MaxEvents 20
```



Step 5: Count failed login events and generate an alert

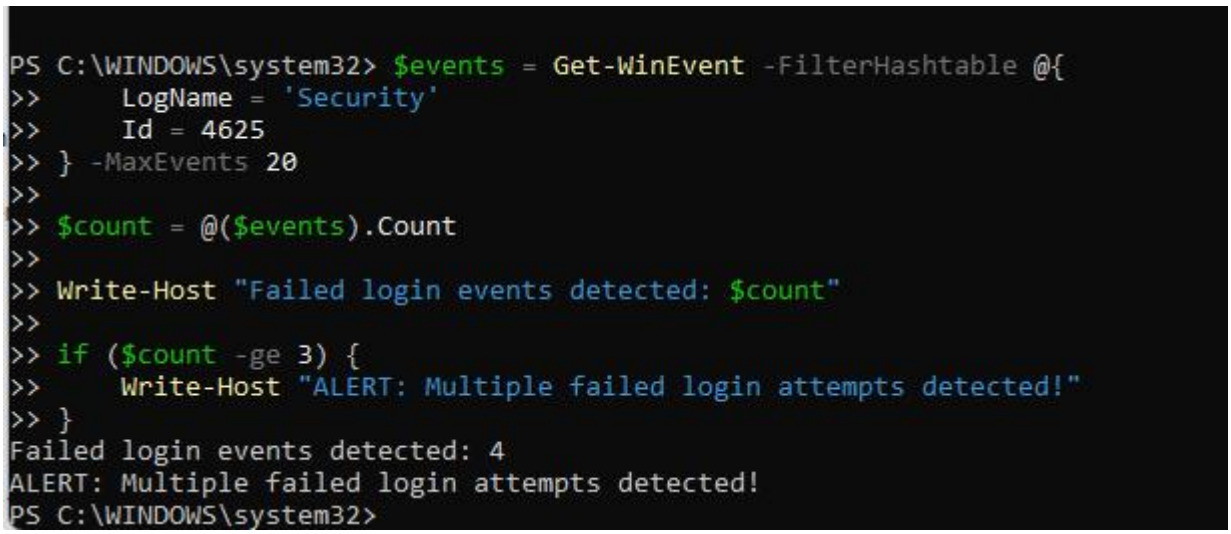
Copy and run the complete block:

```
$events = Get-WinEvent -FilterHashtable @{
    LogName = 'Security'
    Id = 4625
} -MaxEvents 20

$count = @($events).Count

Write-Host "Failed login events detected: $count"

if ($count -ge 3) {
    Write-Host "ALERT: Multiple failed login attempts detected!"
}
```



Step 6: Create a folder for forensic evidence

We will use a simple folder path on the C drive.

Run:

New-Item -ItemType Directory -Path "C:\DMF\Exp3" -Force

Step 7: Export the failed login events to CSV

Run the following command:

Get-WinEvent -FilterHashtable @{

LogName = 'Security'

Id = 4625

} -MaxEvents 20 |

Select-Object TimeCreated, Id, ProviderName, MachineName, Message |

Export-Csv "C:\DMF\Exp3\forensic_events.csv" -NoTypeInformation

The file will be saved here:

`C:\DMF\Exp3\forensic_events.csv`

This file contains the timestamps, event IDs, computer information, and event messages.

OUTPUT

TimeCreated	Id	ProviderName	MachineName	Message
21-10-2026 15:42	4625	Microsoft-Windows-Security-Auditing	SECE	An account failed to log on. Subject: Security ID:S-1-5-18 Account Name:SECE$ Account Domain:WORKGROUP Logon ID:0x3E7 Logon Type:2 Account For Which Logon Failed: Security ID:S-1-0-0 Account Name:- Account Domain:- Failure Information: Failure Reason:An Error occured during Logon. Status:0xC000006D Sub Status:0xC0000380 Process Information: Caller Process ID:0x8a4 Caller Process Name:C:\Windows\System32\svchost.exe Network Information: Workstation Name:- Source Network Address:127.0.0.1